NEXORA CARE FLOW

Appointment Records – Feature Engineering Report

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1. Executive Summary

Feature engineering represents the third stage of the Nexora Care Flow appointment forecasting project. The purpose of this stage was to transform the cleaned appointment records into a structured dataset that can support the prediction of future appointment demand at clinic level.

The original dataset contains 129,353 appointment records across 16 variables. The records include information about clinics, appointment dates and times, appointment types, booking channels, holidays, seasonal classifications, and other operational characteristics.

For business stakeholders, the key purpose of feature engineering is straightforward: to turn historical appointment information into useful indicators of future demand. Rather than looking at appointments only as individual historical transactions, the analysis reorganised the information to show how busy each clinic was on a daily basis and whether demand followed identifiable patterns.

The process created daily appointment volumes for each clinic and added calendar, holiday, seasonal, historical-demand, and recent-average information. These features provide the foundation for the next stage, where the forecasting model will use historical patterns to estimate future appointment demand.

2.0 Purpose of Feature Engineering

The Nexora Care Flow project aims to help clinics move from reactive staffing decisions towards proactive planning. Administrators currently have limited forward visibility of appointment demand and may therefore find it difficult to determine how much staffing and capacity will be required in advance.

The feature-engineering stage addresses this by preparing the historical data in a way that allows patterns in demand to be identified and subsequently used for forecasting.

The main objectives were to:

- Create a clear daily measure of appointment demand for each clinic.
- Capture recurring calendar patterns.
- Retain information about holidays and seasonal periods.

- Capture recent historical appointment activity.
- Create measures that show the average level of recent demand.
- Prepare a consistent dataset for the forecasting stage.
- Ensure that historical information is used appropriately without allowing future information to influence past observations.

2.1 Feature Engineering Approach

2.1.1 Clinic-Level Daily Demand

The individual appointment records were grouped by **ClinicID, ClinicName, and date**. The number of appointments within each group was calculated as `AppointmentCount`.

This provides a daily measure of demand for each clinic rather than treating every appointment as an independent observation. The approach is directly aligned with the project objective of producing **clinic-level appointment forecasts**.

This structure will allow future forecasts to identify whether a particular clinic is expected to experience higher or lower appointment demand.

2.2 Calendar Features

Calendar information was extracted from the appointment date to capture recurring patterns in demand.

The following features were created:

- Day of week
- Day name
- Day of the month
- Month
- Month name
- Quarter
- Year
- Week of the year
- Weekend indicator

These features provide context around when appointments occur and allow the forecasting analysis to identify recurring patterns, such as differences in demand between weekdays and weekends or changes across different periods of the year.

2.3 Holiday and Seasonal Information

Existing holiday and seasonal information was retained during the transformation.

The `IsHoliday` and `HolidayName` fields allow appointment demand on holidays to be compared with normal periods. This is relevant because clinic activity may change around holidays due to differences in patient behaviour, operating arrangements, or appointment availability.

The `SeasonFlag` field was also retained. The dataset identifies **Standard, Flu Season, and Allergy Season** periods.

Including these indicators allows the forecasting process to assess whether appointment volumes vary during the defined seasonal periods.

3. Historical Demand Features

Historical appointment activity was incorporated into the feature-engineered dataset using lag measures.

The analysis created:

Feature	Purpose
Lag 1	Captures recent appointment activity
Lag 7	Captures activity from approximately one week earlier
Lag 14	Captures activity from approximately two weeks earlier
Lag 28	Captures activity from approximately four weeks earlier

These measures provide historical context for each clinic. For example, recent appointment activity may provide useful information when estimating the likely level of demand in an upcoming period.

The lag calculations were performed separately for each clinic and after the data had been placed in chronological order.

4. Rolling Average Features

In addition to individual historical observations, rolling averages were created to provide a broader view of recent appointment demand.

Three measures were developed:

- **7-period rolling average**
- **14-period rolling average**
- **28-period rolling average**

These averages help smooth short-term fluctuations and provide an indication of the underlying level of recent demand.

For example, a single unusually busy day does not necessarily indicate a sustained increase in demand. A rolling average allows the analysis to consider activity across several previous periods.

Importantly, the calculations were based on previous observations rather than including the current appointment volume. This helps ensure that information from the period being predicted is not inadvertently used to create its own forecasting features.

5. Chronological Data Preparation

The clinic-level records were sorted by **ClinicID and Date** before historical features were created.

This was an important step because the project is based on time-series forecasting. Historical information must remain in the correct sequence so that previous appointment activity is used to inform later observations.

The approach therefore maintains the distinction between **past information and future information**, providing a more reliable foundation for subsequent forecasting.

6. Resulting Dataset

The final feature-engineered dataset brings together the key information required for the forecasting stage.

Main feature groups

Clinic information

- ClinicID
- ClinicName

Appointment demand

- Daily AppointmentCount

Calendar information

- Day
- Day of week
- Month
- Quarter
- Year
- Week of year
- Weekend indicator

Holiday information

- IsHoliday
- HolidayName

Seasonal information

- SeasonFlag

Historical demand

- Lag 1
- Lag 7
- Lag 14
- Lag 28

Recent demand

- 7-period rolling average
- 14-period rolling average
- 28-period rolling average

The completed dataset was saved as **`Nexora_Feature_Engineered_Data.csv`** for use in the forecasting stage.

7. Business Value

The main business value of this phase is the conversion of historical appointment records into information that can support forward-looking operational planning.

Previously, the appointment data primarily described what had already happened. Following feature engineering, the dataset provides information about:

- How busy each clinic has been.
- Recent appointment activity.
- Recurring calendar patterns.
- Holiday periods.
- Seasonal periods.
- Changes in demand over recent weeks.

This creates the foundation for a forecasting model that can estimate future appointment volumes.

For clinic administrators, reliable forecasts could eventually support better decisions about **staffing, room allocation, workload management, and preparation for periods of increased demand**.

8. Limitations

The feature-engineering stage does not by itself demonstrate that the created variables will produce accurate forecasts. Their usefulness will need to be assessed during model development and validation.

The current analysis is also based on the historical data available within the pilot dataset. As Nexora Care Flow collects additional appointment records, the forecasting process can potentially benefit from a longer history and more observations.

Therefore, the current output should be viewed as the foundation for forecasting rather than the final forecasting solution.

9. Conclusion

The feature-engineering phase successfully transformed the Nexora Care Flow appointment records into a structured **clinic-level daily dataset** suitable for forecasting.

Calendar, holiday, and seasonal information was incorporated to capture important patterns, while lag and rolling-average measures were created to represent recent and historical appointment demand. The data was organised chronologically to ensure that historical information was used appropriately.

The resulting dataset provides the necessary foundation for the next project phase: developing and validating a forecasting model that can estimate future appointment demand and provide useful guidance for proactive staffing and resource planning.

Ultimately, this work supports Nexora Care Flow's wider objective of helping clinics move away from reactive, intuition-based planning towards practical, data-driven operational decision-making.